

An improved X-ray image diagnosis method for COVID-19 pneumonia on a lightweight neural network embedded device

Ziyang Zhang*

Chengdu University of Technology,
China
ziyangzhang02@outlook.com

Yingjie Ma[†]

Chengdu University of Technology,
China
muringj2248@outlook.com~

Keran Li[‡]

Chengdu University of Technology,
China
keranli98@outlook.com

ABSTRACT

With the continuous variation and evolution of COVID-19, Deep learning-based X-Ray image classification models of COVID-19 have been emerged a lot and achieved high accuracy rates. However, the current mainstream neural network algorithm generally has too many parameters, complex models and high computational complexity. It is usually deployed on high-performance computing servers, which is not conducive to embedded deployment and real-time computing. In order to better enable the neural network to play its role on embedded devices without reducing the classification accuracy of the model, this paper proposes MECNet (MobileExprCovidNet) by redesigning the network structure and transforming the attention mechanism with cutting-edge knowledge, which ensures the accuracy rate compared to MobileNetV3 has been greatly improved. Compared with networks with larger number of relative parameters and higher computational complexity, MECNet does not have significant disadvantages in terms of accuracy and recall. MECNet is more suitable for implementation on various embedded architecture compared to the original MobileNetV3 network structure. Experimentally, the proposed network improves 4%, 5%, and 5% in accuracy, precision, and recall compared to the base model MobileNetV3 for the publicly accessible COVID-19 Radiography Database dataset image class. Outperformed many metrics on embedded device experiments GhostNet, ResNet50, MobileNetV3. The accuracy, precision, and recall rates are 92%, 90%, and 91%, while only 73MFLOPs are required, and the number of parameters is only 0.85M, which meets our requirement for deployment on embedded machines. The Chest X-ray dataset is used to further verify the generalization ability of the network, and Score-CAM is used to verify its effectiveness. The experiments demonstrate that the MECNet proposed in this paper has a good classification effect for COVID pneumonia X-rays. Finally, the model was deployed to an embedded device, and the processing speed was measured to

reach 183.3ms, and the accuracy rate reached 89.7%, which directly proves the usability and efficiency of the model on the embedded device.

CCS CONCEPTS

• **Computer systems organization** → Embedded systems..

KEYWORDS

low computational complexity, X-Ray image, COVID-19

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1 INTRODUCTION

In the onset stage of the COVID-19 outbreak, the most common method for detecting COVID-19 during the pandemic has been the reverse transcription polymerase chain reaction (RT-PCR) test kits, but limitations in sampling techniques, uncontrollable factors in transit, kit quality issues and evolution of COVID virus can all lead to a high probability of false-positive nucleic acid tests. The main means of analysis and diagnosis of pneumonia is chest X-ray [12]. And because of the uniqueness, objectivity, and simplicity of medical X-ray images of COVID pneumonia, chest X-ray is now one of the most important tools for confirming the diagnosis of COVID pneumonia and is an important component of follow-up and early detection. The efforts of many front-line researchers and medical practitioners have led to the disclosure of chest X-ray image data of patients who have been diagnosed with COVID-19, making the application of deep learning, a data-driven algorithm, for the diagnosis of COVID pneumonia increasingly widespread. At the same time, with the spread of the COVID epidemic, the epidemic has also spread from the city to the countryside, covering a wider and wider area. One rather serious problem in the undeveloped area is the lack of medical and nursing resources, financial capacity and manpower shortage in rural hospitals. Very often, rural hospitals are unable to spare the manpower to diagnose patients, and many patients return home with further cross-infection, resulting in an epidemic that cannot be effectively suppressed, and the condition of the patients and the medical crowding situation only gets worse.

At this stage, researchers have designed many deep learning diagnostic models about COVID pneumonia and achieved high accuracy rates in the medical field.

Andrew Ng's team proposed CheXNeXt's X-ray diagnostic algorithm, a model that can diagnose 14 diseases, and in 10 of them

*Ziyang Zhang, male, focus on computer vision and high performance computing. Email: ziyangzhang02@outlook.com

[†]Yinjie Ma, male, focus on computer vision and AI related region. Email: muringj2248@outlook.com

[‡]Keran Li, male, focus on computer vision for geography and geography related region. Email: keranli98@outlook.com

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the AI performed on par with human radiologists, and in one more than humans. Debaditya Shome, T. Kar et al. proposed a method for medical diagnosis using Vision Transformer, which achieved excellent results in the precision and reliability [15]. Daniel Kienzle et al. proposed a diagnostic method using ConvNext that also achieved satisfactory results [9]. The above progress shows that the medical diagnosis method using machine learning has obvious advantages. However, the networks used by researchers at this stage generally have more parameters and larger models than really embedded system capacity and are mostly deployed in high-performance computing servers. As applied to many scenarios of embedded deployment and use there are high hardware requirements and large hardware costs. Coupled with the large population base for pneumonia diagnosis in the current environment and the relatively large demand for use the high cost has hindered the practical application of research results and the industrial application process. Therefore, we urgently need a lightweight, low-cost, efficient, real-time diagnostic algorithm that can be deployed on mobile devices to assist rural doctors to diagnose COVID pneumonia quickly and easily on simple mobile devices.

Inspired by the above-mentioned ideas and studies, and the problem of improving the network classification accuracy on radiographic images of COVID-related lung inflammation through the size of network parameters and combining global and location information, this paper designs two improvements for adapting and effectively solving the above problem. The first proposed change SE mechanism (Squeeze-and-Excitation attention) [7] to CA (Coordinate Attention) mechanism [4]. This move takes advantage of the fact that the CA attention mechanism is sensitive to coordinates and position while adjusting the channel mechanism to capture the extremely important position information in the X-ray of the COVID pneumonia without massively increasing the computational effort and avoiding the loss of position information caused by the SE mechanism. CA simultaneously enhances the network to obtain location information without adding significant computational effort relative to the SE mechanism, ensuring low prediction latency on the embedded platform. Secondly, the complexity of the module task is taken into account by using Neural Architecture Search (NAS) method to search the backbone network parameters. This approach allows for fine-tuning the network's parameter scale without substantially compromising accuracy, which greatly reduces the total network complexity and improves the speed while ensuring that the ability of correctness and generalization degree will not be significantly degraded to make the network perform better on embedded devices. The above improvements lay the foundation for the realization of a low-resource and feature-rich diagnostic system for embedded deployments.

2 RELATED WORK

Nowadays, the combination of CV and medical image collection had been flourishing. The number of papers on static medical imaging using computer vision technology has increased from hundreds to thousands. Radiology, pathology, ophthalmology, dermatology and other fields have received extensive attention.

CNN convolution neural network plays a significant role in the field of medical image and computer vision. [10] It is brilliant in

both medical image segmentation and disease image classification tasks.

Discussions by Esteva, Andre and Chou et al [2], convolution neural network has been brilliant in medical imaging, pathology, dermatology and other disciplines, and has become a powerful force for doctors.

The diagnostic accuracy of some neural networks exceeds that of physicians, and it takes much less time to compute. At the same time, whether discussed by JannisBorn, David Beymer et al.[1] or by Feng Shi, Jun Wang, Jun Shi et al, [14] it can be found that artificial intelligence has now been brilliant in the diagnosis of COVID pneumonia, helping doctors to reduce a large amount of work in daily diagnosis. At the same time, whether Xin Li, Yao Qiang et al. [11] or Rachna Jain, Meenu Gupta, Soham Taneja et al. citeapostolopoulos2020covid which initiated the study further demonstrated the effectiveness of deep learning neural networks as an efficient means of detecting pulmonary disease and COVID pneumonia.

Transformer neural networks are now in vogue, and they also perform exceptionally well in medical tasks. Debaditya Shome, 1 T. Kar et al. proposed a method for medical diagnosis using Vision Transformer, which achieved excellent accuracy. [15]

On the embedded side, lightweight networks such as MobileNetV3[5], GhostNet [3]show their advantages on the embedded and mobile side, and they do not incur a large amount of computational burden with higher accuracy. Andrew Howard, Mark Sandler et al. [6] demonstrate that MobileNetV3 has room for improvement in attention mechanisms for different tasks. Now thank V. Santhosh Kumar Tangudu, etc. scholars efforts mobileNet also achieved satisfactory results on the covid task. [16]

Nevertheless, due to the shortage of resources of embedded devices, the embedded deployment of neural network on general hardware can only achieve a single task. There is still a large research space to integrate its functions as a part of complex tasks.

Comparing with other lung diseases, according to Adam Jacobi, Michael Chung et al. [8], we can find that X-ray findings of COVID pneumonia have a strong positional correlation. So instead of using the original SE mechanism, we use Coordinate Attention (CA) mechanism [4] enhances the network's capacity to extract location information from nearby or distant regions while maintaining its computational efficiency.

Inspired by ConvNext, it is further transformed with lightweight ideas and neural network technologies of the modern frontier to enhance feature extraction capabilities and classification results instead of relying on simple hierarchical stacking. The network structure is simplified to retain only the most critical part of the structure according to the target situation in this paper, and the network parameters are substantially reduced to remove redundant calculations while maintaining basic effects. To achieve a large reduction in the size of the network and the computation. In the case of relatively good network performance, the number of parameters decreased to nearly 10% of MobileNetV3, while the effect is relatively better than that of MobileNetV3, GhostNet and other lightweight networks.

3 DESIGN OF NETWORK

3.1 Coordinate Attention

Humans can find significant and interesting areas in daily life and complex scenes. Driven by the imitation of human visual system in the realm of computer vision field, the attention mechanism of neural network has been proposed and used. The attention mechanism can be seen as a process that dynamically adjust weights based on the characteristics of the input image. It tells the neural network more about "why" and "where". It has been widely used in the realm of computer neural network and has reached a significant achievement in many tasks, including image classification, target detection, etc.

The CA module is located after each module is convolved to effectively obtain the global information, focus on the significant information related to the abnormalities or pathological changes, and suppress the remaining invalid information to enhance the network's classification precision. Channel attention consists of three modules. By introducing normalization instead of full connection layer to model the characteristic relationship between channels, parameters and convolution weights are jointly optimized to determine the competition and cooperation status of neurons. The module focuses on the relationship between different channels, and finally uses GeLU to activate the control attention vector to achieve more accurate recognition. Spatial attention focuses on detecting and capturing detailed spatial features in a specific area or region, the convolution and superposition of three small receptive fields, and the GeLU activation function replaces ReLU, which improves the training stability and accuracy.

Convolutional attention module CA discusses the relationship between the positions of attention modules in detail, infers attention maps along the coordinate sequence of channels and images, and has almost no more computational overhead compared with the traditional SE mechanism, with fewer parameters, which meets our needs to deploy on embedded devices.

CA module consists of channel attention (GCT) and long-distance dependent attention (LISA). More precisely, when provided parameter X , we use the convolution kernels $(H, 1)$ and $(1, W)$ of two spatial ranges to perform convolution and coding operations in both the horizontal and vertical orientations respectively. So, the output of the c th channel with height H is able to formally describe as

$$z_c^h(h) = \frac{1}{W} \sum_{0 \leq i \leq W} X_c(h, i) \quad (1)$$

Likewise, output of channel c with a width of w , can also be expressed as

$$z_c^w(w) = \frac{1}{H} \sum_{0 \leq i \leq H} X_c(j, w) \quad (2)$$

The aforementioned two operations produce features along two spatial orientations respectively. Compared with SE squeezing into feature vectors, these transformations can let our module to obtain the long-distance reliance of one direction while maintaining precise position information of our spatial direction along the other direction. This will help the network find the required information in the location related tasks more accurately.

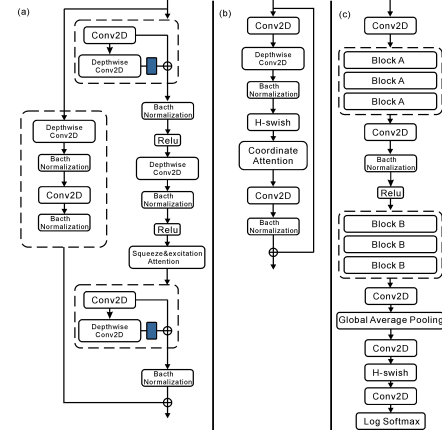


Figure 1: Schematic diagram of neural network

As mentioned previously, Formula (4) and Formula (5) can open a global receptive field while decoding position information. Then we connect the aggregation feature maps generated based on Formula 4 and Formula 5, and subsequently transmit them to a common 1×1 convolution layer to get the next formula

$$f = \delta \left(F_1 \left(\left[z^h, z^w \right] \right) \right) \quad (3)$$

Where $[x, y]$ denotes the Concatenation operation in the spatial dimension, δ symbolizes a nonlinear activation transformation, and $f^h \in \mathbb{R}^{C/r \times (H+W)}$ represents an intermediary feature graph, representing positional is be captured along the X-Axis and Y-Axis. R here is just like the SE block, representing the reduction rate. Then we segment f to two independent tensors $f^h \in \mathbb{R}^{C/r \times H}$ and $f^w \in \mathbb{R}^{C/r \times W}$ along the spatial dimension. There are also have 1×1 convolution kernels F_h and F_w are using for transforming F_h and F_w makes the quantity of channels the same for parameter X , and the result is

$$g^h = \sigma \left(F_h \left(f^h \right) \right) \quad (4)$$

$$g^w = \sigma \left(F_w \left(f^w \right) \right) \quad (5)$$

Remember that σ is the sigmoid transform. For minimize the model complexity, we use a suitable reduction ratio to decrease the channel count off. The outputs g^h and g^w are expanded and then used as additional attention weights. Here, the result of the attention associated with the coordinates is formally describe as

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j) \quad (6)$$

3.2 MECNet Framework

MECNet comprehensively considered the model parameter quantity, calculation quantity and accuracy, used the advanced experience of mainstream lightweight networks such as MobileNetV3 and GhostNet, combined with the relevant cutting-edge knowledge achievements of attention mechanism to design the network, used 5×5 depth separable convolutional kernel to effectively extract features, as well as linear bottleneck structure and reverse residual structure to avoid information loss, and properly used the association and repetitiveness between feature maps, It also adopts

Table 1: Dataset number of sample images divided

dataset	types	Count of CXR's	Training	Validation	Test
COVID-19	COVID	3615	2531	723	361
Radiography	normal	10192	7131	2038	1020
Database	lung_opacity	6012	4208	1202	602
	Viral Pneumonia	1345	941	269	136

Table 2: Dataset number of sample images divided

dataset	types	Count of CXR's	Training	Validation	Test
Chest X-ray	Covid	576	403	115	58
	normal	1583	1108	316	159
	Pneumonia	4273	2991	854	428

the Coordinate Attention mechanism with better information capture capability and lower computational overhead to enhance the precision of the network. When the network has achieved good results, the model parameters are greatly reduced and NAS technology is used to adjust the model parameters to achieve a super lightweight model that can be used for deployment of low-cost embedded devices.

The network are shown as Figure 1, it can be divided into A-D four parts:

Part A: The input image is rotated and cropped to 175 * 175 without using other data enhancement methods. Reduce the image height and width and increase the channel size through convolution operation to obtain the feature map $F_1 \in \mathbb{R}^{88 \times 88 \times 96}$. Part B: The feature map enters the main structure. First, it is subjected to depth convolution (DW), normalized by Batch Normalization (BN), and then preliminarily processed by Block A, and activated by SE and HS functions to capture key feature information to obtain the feature map F_2 . Part C: After get F_2 , Block B further uses the attention mechanism for focus on the valuable information, and further quickly extracts the features to obtain F_3 . Part D: F_3 is normalized by Batch Normalization (BN), the feature map size is reduced by 2 x 2 convolution, and the channel size is increased. Enter the Block structure, extract feature information through depth convolution (DW), and finally obtain the output $F_4 \in \mathbb{R}^{11 \times 11 \times 96}$. F_4 is normalized by BN, pooled and convolved, and finally logged_Softmax classification to get the final output results.

4 EXPERIMENTAL METHOD

4.1 Datasets

Our dataset is come from the COVID-19 Radiography Database Version 5[17]. The experiment uses 3616 images of COVID pneumonia, 6012 images of lung turbidity, and 8851 normal images. In the experiment, 70% of the dataset are using for training, 20% testing, and 10% validation to avoid overfitting. Our dataset comes from Kaggle public dataset Chest X-ray COVID Pneumonia dataset. [13] Number of dataset partition is displayed by Table 1.

And our dataset comes from Kaggle public dataset Chest X-ray COVID Pneumonia dataset. Number of dataset partition is displayed by Table 2. [13]

4.2 Evaluation standard

In order to measure the effectiveness of the proposed MECNet network, Accuracy, Precision, Recall, and F1-score are used as evaluation indexes, where TP represents the count of correctly predicted positive samples, TN represents the count of correctly predicted negative samples, FP represents the count of incorrectly predicted positive samples, and FN represents the count of incorrectly predicted negative samples, and the calculation formula is as follows.

$$accuracy = \frac{TP + TN}{TP + TN + EP + FN} \quad (7)$$

$$recall = \frac{TP}{TP + FN} \quad (8)$$

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

$$F_1 = 2 \cdot \frac{precision \cdot recall}{precision + recall} \quad (10)$$

4.3 Experimental platform

On the embedding system testing side, we chose Raspberry Pi 3B as our experimental platform, which has 1G RAM and 64-bit 1.2GHz quad-core ARM Cortex-A53, as illustrated in Figure 2, and we used pure CPU for inference, which is closer to most embedded platforms and has universal applicability.

5 EXPERIMENT

5.1 Experiment result

First, we compared the MCENet network model with the model size and parameter quantity of mainstream network models like ResNet50, MobileNetv3, and GhostNet to prove the comprehensive applicability of the MCENet model in this application scenario, displayed in the Table 3.

Then conduct one hundred rounds of training on MCENet and other models. To visually demonstrate the outcomes and efficacy of the trial, the experiment draws the accuracy change curve of

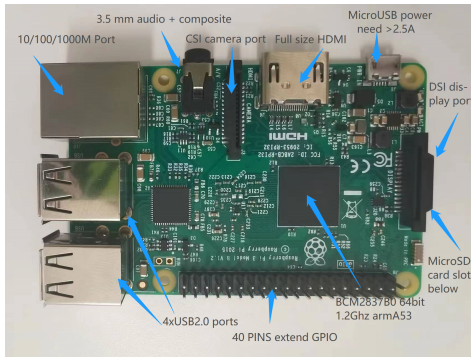


Figure 2: Raspberry Pi 3B

Table 3: Comparison of mainstream network and ours (MCENet)

models	mParams	mFLOPs
ResNet50	25.6	4857
MobileNetv3	5.4	80
GhostNet	7.3	218
MCENet(Ours)	2.59	73

Table 4: MCENet's indexes on dataset I

Type of illness	Pre%	Rec%	F1-score%
Covid	94	93	94
Normal	88	94	86
Lung_opacity	90	92	91
Viral Pneumonia	96	90	93

Table 5: Other's indexes on dataset I

models	Pre%	Rec%	F1-score%
MobileNetv2	80	82	80
MobileNetv3 (baseline)	88	85	86
GhostNet	89	89	89
MCENet(Ours)	92↑	90↑	91↑

MCENet and the confusion matrix on test set. It's shown as the Figure 3 that the correct rate rises rapidly after about 10 rounds of training, and the curve gradually stabilizes after 20 rounds until 100 rounds.

At the same time, Table 4's experimental results demonstrate the strong classification performance achieved by MCENet on the COVID-19 Radiography Databases dataset, which proves the effectiveness of the network. Then, the effectiveness of the model was verified by testing the dataset with other mainstream mobile network models

for horizontal and vertical comparisons. The validity of the MCENet was also demonstrated by conducting comparisons as shown in Table 5.

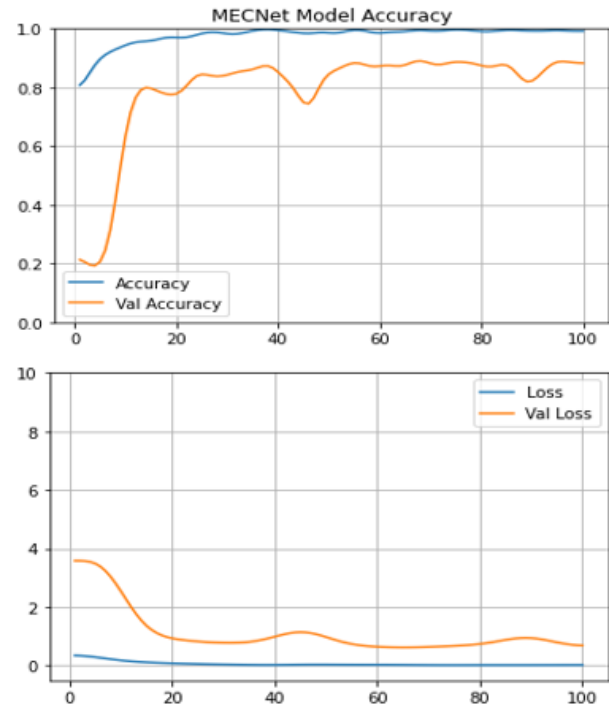


Figure 3: Training process

Finally, to validate the efficacy of our method, the experiments use the confidence score visualization technique (Score-CAM), which the visualization highlights regions of great interest to the model during network learning, meaning that the feature maps generated in the convolutional layers contain information that helps to identify focal regions. As the displayed in Figure 4, which illustrates the feature maps in our network, the images show that our network is able to accurately and efficiently learn the focal region on the chest X-ray images of the COVID-19.

5.2 Ablation experiment

In order to verify the effect of each module of MCENet, we designed an ablation experiment for verification. First of all, we tested the model effect of mobileNetv3, and the measured indicators were 88%Pre, 85%Rec, and 86%F1. Then we tested the effect of the model with only Block A added, and its indicators were the same as the baseline Pre and increased by 5%Rec and 2%F1-score. Then we tested the effect of the model with only Block B, and its indicators were the same as the baseline Pre and increased by 3%Rec and 2%F1. Finally, we tested the effect of the model including both Block A and Block B, and its indicators increased by 4%Pre, 5%Rec, and 5%F1 year-on-year. The final results are shown in the Table 6. The experiments fully illustrate the effectiveness and necessity of each structure of our proposed model.

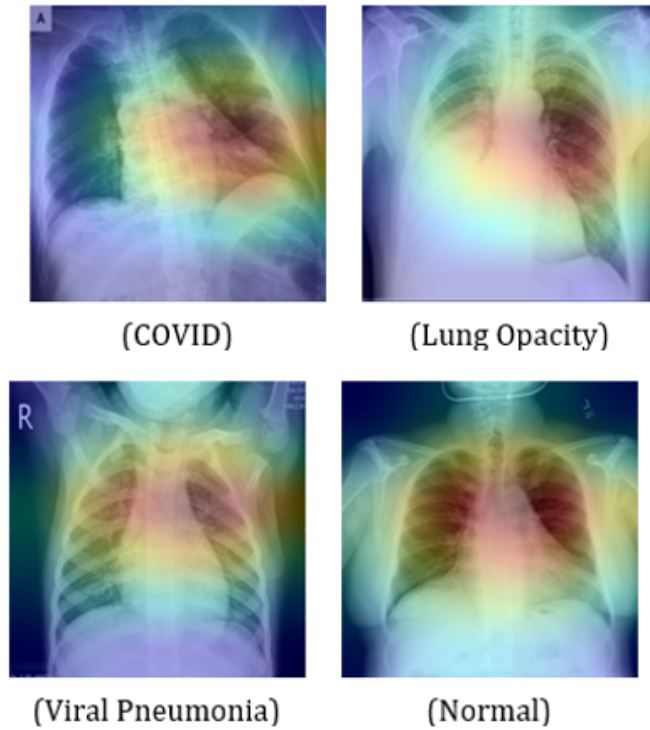


Figure 4: Score-CAM HotMap

Table 6: Comparison of ablation experimental indexes

models	Pre%	Rec%	F1-score%
MobileNetV3(baseline)	88	85	86
Res50(BigParams)	90	90	90
Block A only	88	90	89
Block B only	88	88	88
Block A and Block B Both	92↑	90↑	91↑

5.3 Confusion matrix and Generalized Performance Testing

We again present the prediction result situation in the network more visually as a chaos matrix. Also, generalization performance is always an important metric in neural network systems, and we again use the data set shown in Table2 to make predictions, and the final confusion matrix and situation is as the Figure 5 .

5.4 Embedding system deploying and delay

The embedding experiment uses a standard Raspberry Pi 3 to simulate the embedded deployment environment, while using TF-lite to ensure that the overhead of the part other than the computation is low and avoid causing interference. The embedding experiment focuses on three indicators, 1. the latency of each judgment of a picture 2. the accuracy of each picture 3. the size of the model. The size of the model most of the time directly determines the running speed of the model, and also avoids excessive memory operations

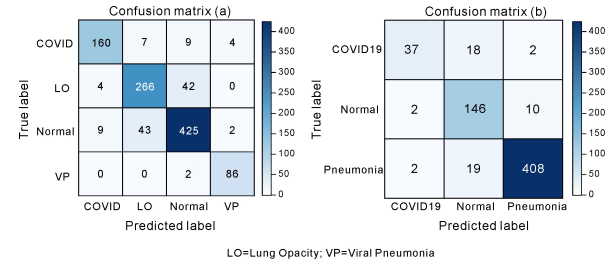


Figure 5: Confusion matrix

Table 7: Comparison of models indexes

models	Delay	Acc%	Size
ResNet50	1439.4ms	90.4%	97.4MB
MobileNetv3	335.4ms	87.2%	12.2MB
MCENet (Ours)	183.3ms	89.7%	2.2MB

and computational slowdowns due to overly large computational maps. In the comparison test here, we use ResNet50, the original MobileNetV3 and the MCENet proposed in the article to compare the latency, accuracy, and model size after deployment as Table 7.

According to the above experiments and results, the size of our MCENet model is only nearly 1/50th of the size of ResNet50, while various index of our MCENet model is qualitatively enhanced and quantitatively surpassed compared to the baseline MobileNetV3, which is a significant improvement, and is sufficient to show that our proposed idea is correct and effective.

6 CONCLUSION

In this research, we propose an extremely lightweight network MCENet that guarantees correctness and generalization performance while guaranteeing a smaller model and computation that can be easily deployed on any platform. To the best of our knowledge, this is one of the smallest models in terms of COVID pneumonia diagnosis, about 2.2 MB. The algorithm proceeds through the following stages: 1. Acquire the image 2. Load and normalize the image data 3. Remove zero offset 4. Rescale to dimensions of 175x175 pixels 5. Enter the network operation 6. The network algorithm gives the result. This network algorithm is deployed on top of Raspberry Pi 3 to verify that it can guarantee fast computation speed and high correctness and generalization performance with low power consumption, extremely limited storage, and computational power. This study raises new questions about how to extract features more efficiently in small-scale classification tasks (with a small number of classifications) by modern advanced techniques, which significantly reduce the number of parameters while ensuring no excessive fluctuations in correctness compared to other lightweight networks. Although customization through parameter reduction is a good choice, can we also consider leveraging the redundant parameters of the standard model with cutting-edge techniques to achieve more effective standardization and efficient deployment?

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